

ANALYSTLAB AFRICA

Data Science Internship Programme



Employee Attrition Analysis Dataset Inspection Report

By

Ogheneughwe Godwin AKISE

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Dataset Inspection Report Submission

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1. Dataset Dimensions (Shape)

Rows: 1470

Columns: 35

2. Column Names

The dataset contains 35 columns, including: Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, EmployeeCount, EmployeeNumber, EnvironmentSatisfaction, Gender, HourlyRate, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, MonthlyIncome, MonthlyRate, NumCompaniesWorked, Over18, OverTime, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StandardHours, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, and YearsWithCurrManager.

3. Data Types

The dataset comprises both numerical (int64) and categorical (object) data types.

Numerical Columns (Examples): Age, DailyRate, DistanceFromHome, Education, HourlyRate, MonthlyIncome, TotalWorkingYears, and others.

Categorical Columns (Examples): Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, Over18, and OverTime.

4. Missing Values

There are no missing values in any of the columns. Each column has 1470 non-null entries, matching the total number of rows.

5. Duplicate Values

There are no duplicate rows in the dataset.

6. Unique Values Summary

Constant Columns: EmployeeCount, Over18, and StandardHours each have only one unique value (1, 'Y', and 80, respectively). These columns are constants and provide no variance, making them candidates for removal in modelling.

Unique Identifier: EmployeeNumber has 1470 unique values, serving as a unique identifier for each employee.

Numerical Features: Columns like Age (43 unique values), DailyRate (886), MonthlyIncome (1349), MonthlyRate (1427), and TotalWorkingYears (40) show a wide range of unique values, as expected for numerical features.

Categorical Features: Columns such as Attrition (2), BusinessTravel (3), Department (3), EducationField (6), Gender (2), JobRole (9), MaritalStatus (3), OverTime (2), PerformanceRating (2), EnvironmentSatisfaction (4), JobInvolvement (4), JobSatisfaction (4), RelationshipSatisfaction (4), StockOptionLevel (4), and WorkLifeBalance (4) have a limited number of unique values, representing distinct categories or ordinal scales.

7. Descriptive Statistics for Numerical Columns

Age: Mean age is approximately 37 years (range 18-60), with a median of 36, indicating a slight right skew.

DailyRate: Average daily rate is 802, ranging from 102 to 1499.

DistanceFromHome: Average distance is about 9 miles, ranging from 1 to 29 miles. 75% of employees live within 14 miles.

MonthlyIncome: Mean income is around 6,503, but the median is 4,919. The mean being higher than the median suggests a right-skewed distribution, with more employees earning lower incomes and fewer high earners.

Satisfaction Metrics (e.g., EnvironmentSatisfaction, JobInvolvement, JobSatisfaction, RelationshipSatisfaction, WorkLifeBalance): These are mostly on 1-4 or 1-5 scales, with means generally around 2.5 to 3, indicating a moderate to high level of satisfaction across the workforce.

Tenure/Experience (e.g., TotalWorkingYears, YearsAtCompany, YearsInCurrentRole): These show varying levels of experience, with medians suggesting many employees have relatively fewer years in these categories.

8. Descriptive Statistics for Categorical Columns

Attrition: 'No' is the most frequent value (1233 occurrences) compared to 'Yes' (237 occurrences), indicating a significant class imbalance in the target variable.

BusinessTravel: 'Travel_Rarely' is the most common with 1043 occurrences.

Department: 'Research & Development' is the largest department (961 employees).

Gender: 'Male' is more frequent (882 employees) than 'Female' (588 employees).

JobRole: 'Sales Executive' is the most common job role (326 employees).

MaritalStatus: 'Married' is the most common status (673 employees).

OverTime: 'No' (no overtime) is more frequent (1054 employees) than 'Yes' (416 employees).

9. Conclusion

The dataset is clean, with no missing or duplicate values. It contains a mix of numerical and categorical features providing a comprehensive view of employee demographics, job-related attributes, and satisfaction levels. Key observations include the class imbalance in 'Attrition', the presence of constant columns that can be dropped, and the skewed distribution of MonthlyIncome. The descriptive statistics highlight areas for further in-depth analysis, particularly concerning age, income, and job roles in relation to attrition.